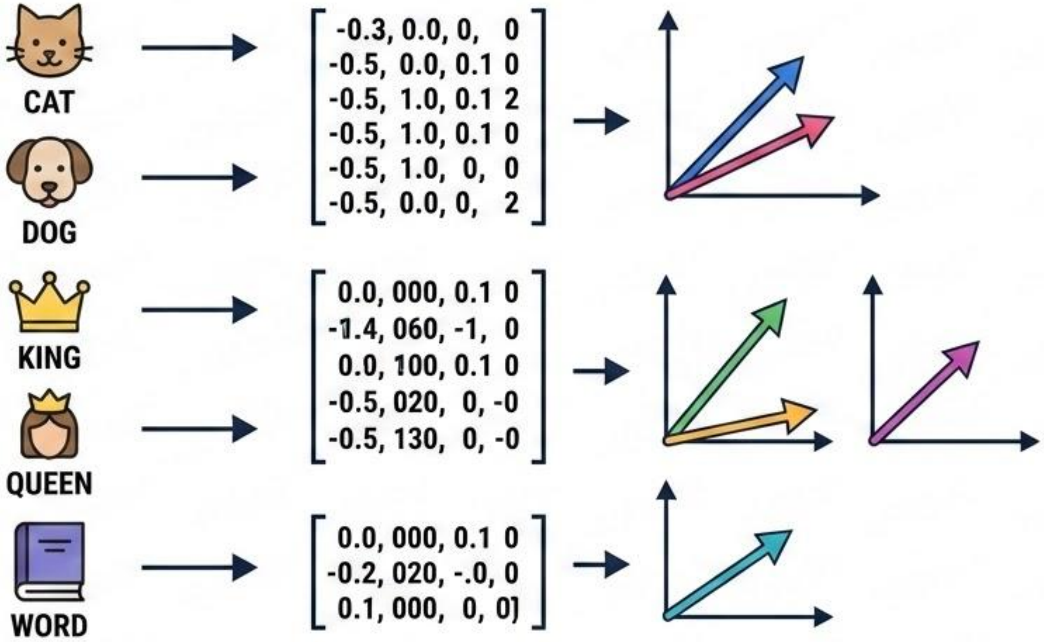
Vector Embeddings



What is a Vector Embedding?

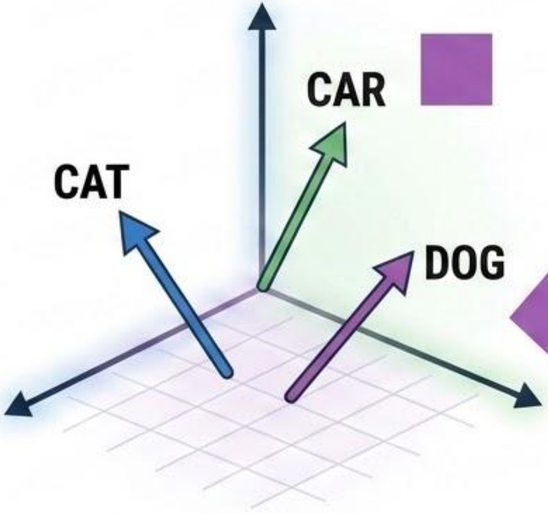
The Secret GPS of Ideas & Objects

A **vector embedding** is just a list of numbers that represents a word, sentence, image, or idea.

CAT

→ [0.23, -0.45, 0.67, ... 0.12]

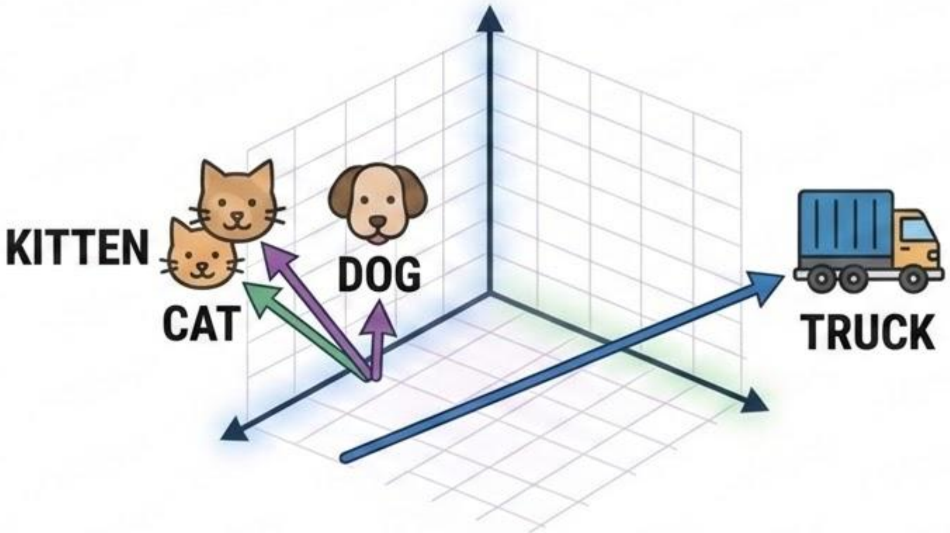
(300+ numbers)



It’s like giving **every thing** its own **secret GPS address** in a giant space.

The Magic Trick

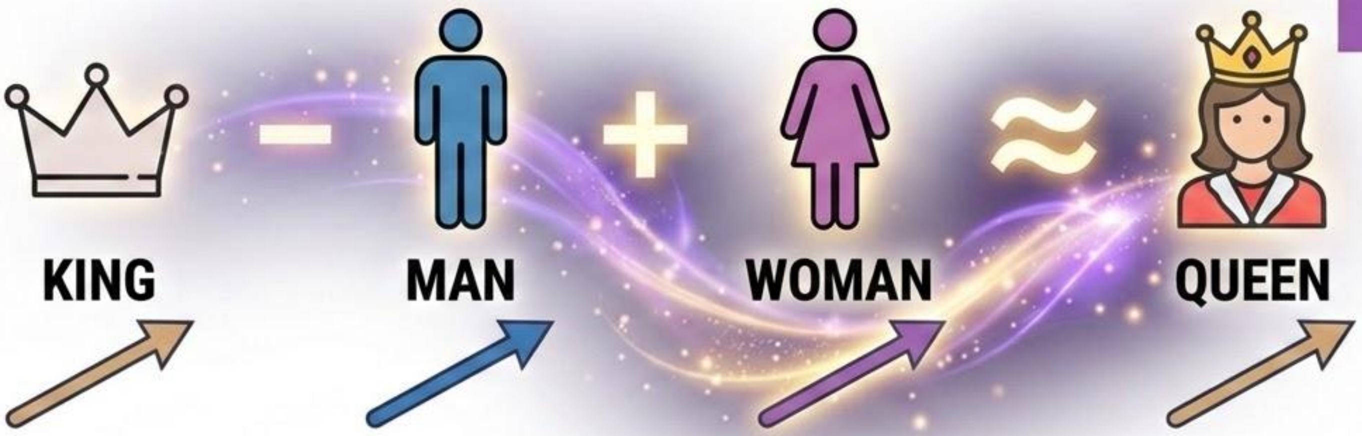
Similar things live close together in this space!



Computers can now **measure meaning** using **distance!**

Famous Vector Math

Math actually works on meaning!



The computer doesn't know **royalty** –
– it just sees patterns in the numbers.

Simple Analogy – The Party

Imagine a huge party hall:

Music Lovers 🎧 **Food Lovers** 🍗 **Introverts**

Embeddings do the same — but with hundreds of **invisible directions!**

Real-World Example – Search

Google & Chatbots use embeddings to understand:

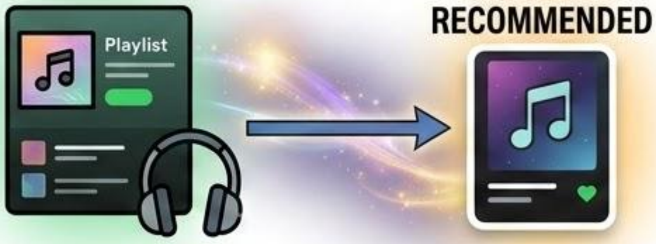
Finds answers even if
words are **different**.

Real-World Example – Recommendations

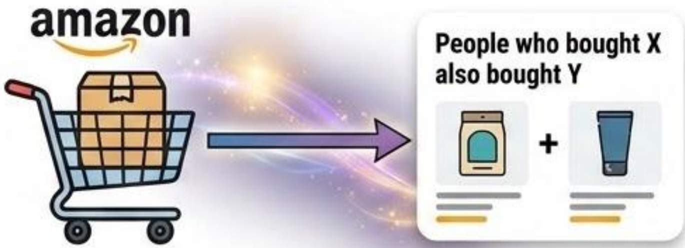
Netflix, **YouTube**, **Amazon**: they turn movies, videos, and products into vectors.

If you liked a movie → they recommend others whose vectors are **closest**.

More Examples – Everyday Life



Spotify: Finds songs with similar "vibe" vectors

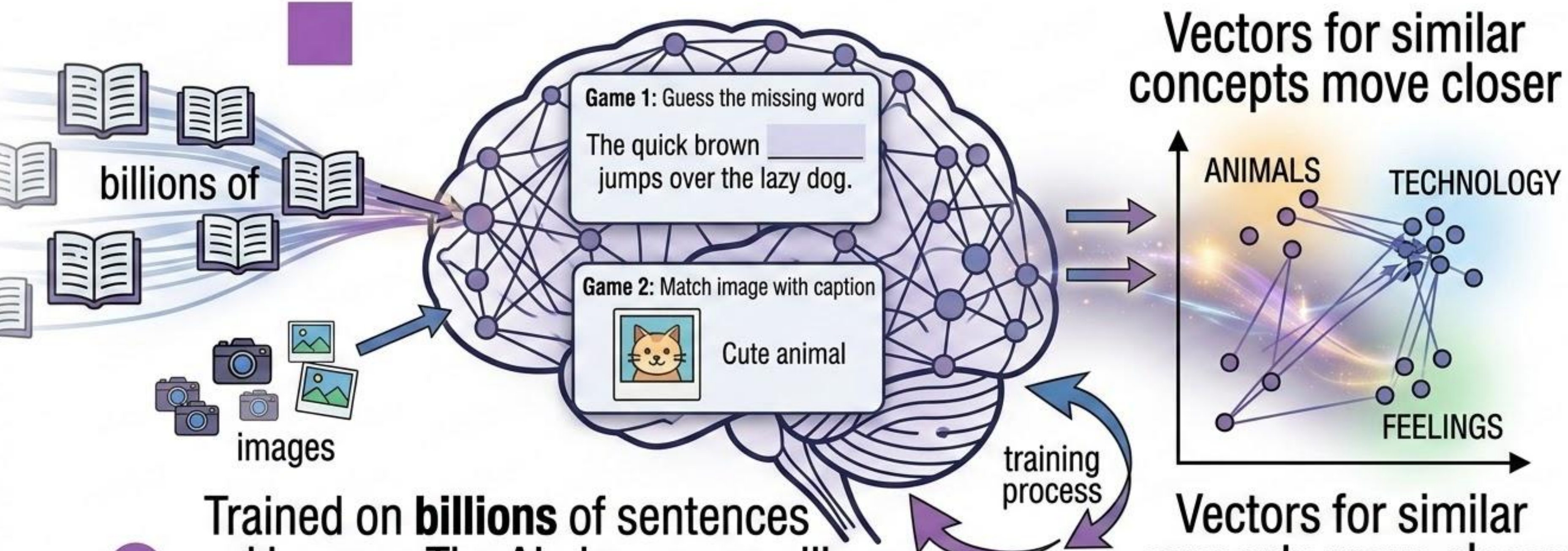


Shopping: "People who bought this also bought..."

Dating apps: Matches people with close personality vectors

Doctors: Compare patient records

How Are Embeddings Created?



Trained on **billions** of sentences and images. The AI plays games like: “Guess the missing word”, “Match image with caption”

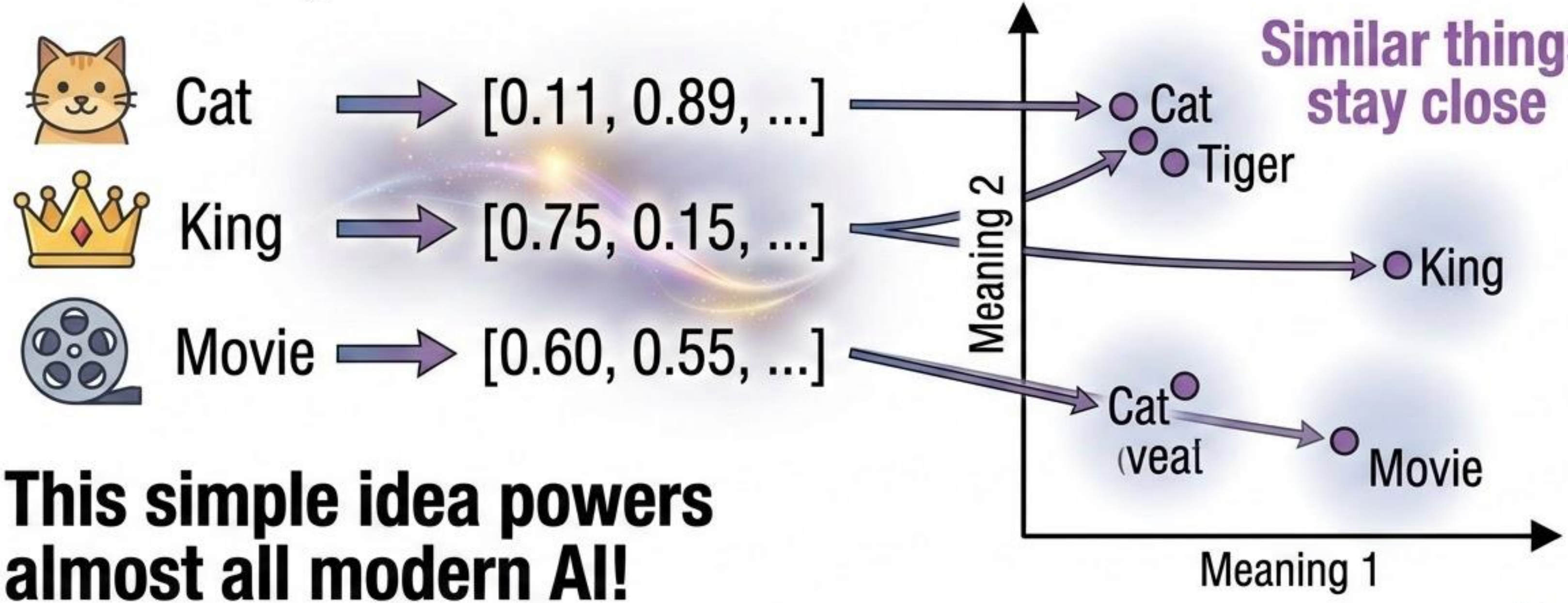
Vectors for similar concepts move closer

Why Is This So Powerful?

- Computers can understand meaning, not just keywords
- Works across languages, images, and text
- Powers modern AI (ChatGPT, Grok, Midjourney, etc.)
- Makes search, recommendations, and chat super smart

Summary

- Vector Embeddings = Lists of numbers that capture meaning
- Similar things = Close vectors



This simple idea powers almost all modern AI!

Thank You!

Questions?

Now you understand how AI turns words and ideas into numbers ✨